

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 YEARLY EXAMINATION PRELIMINARY HSC ASSESSMENT TASK 3 2012

Mathematics Extension 1

General Instructions

- Working time – 90 minutes
- Write using black or blue pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Diagrams are not drawn to scale

Total marks

- Attempt Questions 1- 6
- All questions are of equal value
- Start each question on a new page

Name: _____

Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	TOTAL

QUESTION 1

MARKS

- a) A $(-7, 5)$ and B $(3, 1)$ are two points. Find the co-ordinates of the point P which divides the interval AB externally in the ratio 3:1 2
- b) Solve $\frac{x}{x^2 - 4} \geq 0$ 2
- c) If $x^2 - 7x + 3 = 0$ has two real roots α and β , find the value of
- i) $\alpha + \beta$ 1
 - ii) $\alpha \beta$ 1
 - iii) $\alpha^3 + \beta^3$ 3
- d) If α, β and δ are the roots of $x^3 + 2x^2 + 4x - 5 = 0$ find the value of $(\alpha - 1)(\beta - 1)(\delta - 1)$ 2

QUESTION 2

MARKS

- a) For the function $f(x) = \frac{2x-1}{1-x^2}$
- i) state the behaviour of the curve as x approaches ∞ and $-\infty$ 2
 - ii) write down the equations of any horizontal or vertical asymptotes 2
 - iii) sketch the curve showing all important features 2
- b) i) Express $\cos \theta + \sqrt{3} \sin \theta$ in the form $A \cos(\theta - \alpha)$ where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ 2
- ii) Solve $\cos \theta + \sqrt{3} \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$ 3

QUESTION 3**MARKS**

- a) Find the acute angle to the nearest degree between the vertical line
3
 $x = 2$ and $3x - 2y + 1 = 0$
- b) Derive the equation of the normal at the point $(2ap, ap^2)$
on the parabola $x^2 = 4ay$ 3
- c) Factorise fully the polynomial $P(x) = x^3 - x^2 - 10x - 8$ 3
- d) Solve $|2x - 1| = 4x + 3$ 2

QUESTION 4**MARKS**

- a) i) Show that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ 1
- ii) Hence or otherwise find the exact value of $2 \sin 285^\circ \cos 45^\circ$ 2
- b) Find values of k such that $x^2 + (k + 1)x + 4$ is positive definite 3
- c) Find the area of a regular pentagon of side length 6m
(give answer to the nearest whole number) 3
- d) Find the vertex and focus of the parabola $y^2 = -3x + 6$ 2

QUESTION 5

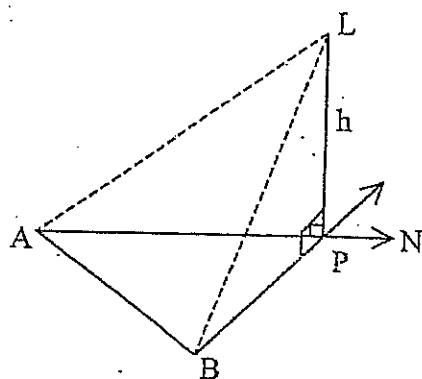
MARKS

- a) One of the roots of $x^2 - (k + 1)x + 2k + 2 = 0$ is twice the other root. Find the value of k 3

- b) i) A monic polynomial is odd, has a single root at $x = -1$ and a double root at $x = 3$. Find its equation, if it is of degree 7 (leave in factorised form) 2

- ii) Sketch this polynomial 1

c)



PL is a vertical pole, h metres high, standing on horizontal ground. PA is a shadow of the pole when the direction of the sun is due north and when the angle of elevation of the sun is 45° . PB is the shadow cast by the pole when the bearing of the sun is 345°T and its angle of elevation is 31° . The distance AB is 15 metres.

- i) Draw a top view sketch and label the information given. 1
- ii) Find the height h of the pole to the nearest metre 4

QUESTION 6

MARKS

a) Find the remainder when the polynomial $P(x) = x^3 - 3x + 1$ is divided by $x^2 - 4$ 2

b) i) What is the equation of the chord of contact to the parabola $x^2 = 8y$, from the external point $T(x_0, y_0)$ 1

ii) If the chord of contact meets the parabola at P and Q, show that the midpoint of PQ is 4

$$M\left(x_0, \frac{x_0^2}{4} - y_0\right)$$

iii) If the point T lies on the line $x + 2y + 5 = 0$, find the locus of M. 4

2012 Ext. 1 Yearly Solutions.

1a)

$B = (3, 1) \quad P(x, y)$

b) $\frac{x}{x^2 - 4} \geq 0$

A.
 $(-7, 5)$

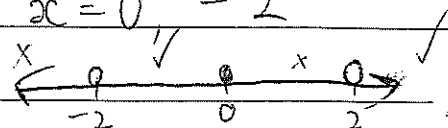
$k: l \equiv -3:1$

Critical pts

$x = 0, \pm 2$

$x = \frac{-3 \times 3 + 1 \times -7}{-3 + 1}$

$y = \frac{-3 \times 1 + 1 \times 5}{-3 + 1}$



$= 8$

$= -1$

$-2 < x \leq 0, x \geq 2$

$(8, -1)$

c) ci) 7

cii) 3

ciii) $\alpha^3 + \beta^3$

$(\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$

$(\alpha + \beta)[(\alpha + \beta)^2 - 2\alpha\beta - \alpha\beta]$

$(\alpha + \beta)[(\alpha + \beta)^2 - 3\alpha\beta]$

$7[7^2 - 3 \times 3]$

$= 280$

d) $x^3 + 2x^2 + 4x - 5 = 0$

$\alpha + \beta + \gamma = -2, \alpha\beta\gamma = 5, \alpha\beta + \alpha\gamma + \beta\gamma = 4$

$(\alpha - 1)(\beta - 1)(\gamma - 1)$

$(\alpha\beta - (\alpha + \beta) + 1)(\gamma - 1)$

$\alpha\beta\gamma - \alpha\beta - \gamma(\alpha + \beta) + \alpha + \beta + \gamma - 1$

$\alpha\beta\gamma - (\alpha\beta + \gamma\alpha + \gamma\beta) + (\alpha + \beta + \gamma) - 1$

$5 - 4 + -2 - 1 = -2$

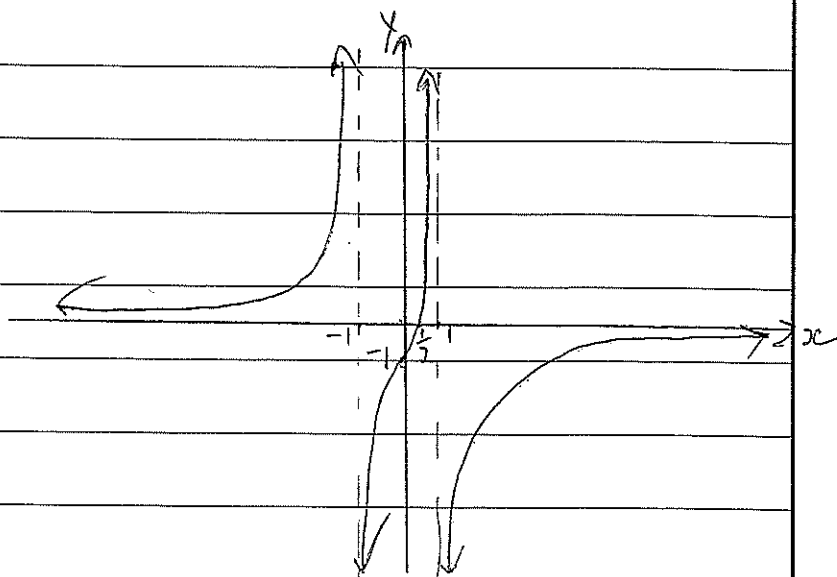
2a) ci) $f(x) = \frac{2x-1}{1-x^2}$

As $x \rightarrow \infty, f(x) \rightarrow 0 \uparrow$

As $x \rightarrow -\infty, f(x) \rightarrow 0 \downarrow$

cii) $x = \pm 1$ vertical

$y = 0$ horizontal



$$b) i) A = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\angle = \tan^{-1} \sqrt{3} \therefore \angle = \frac{\pi}{3}$$

$$c) ii) 2 \cos \left(\theta - \frac{\pi}{3} \right) = 1$$

$$\cos \left(\theta - \frac{\pi}{3} \right) = \frac{1}{2}$$

$$\theta - \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, -\frac{\pi}{3}$$

$$\theta = 0, \frac{2\pi}{3}, 2\pi$$

3a) Use $m=0$ (complementary $\angle$ for $x=2$)

$$3x - 2y + 1 = 0$$

$$2y = 3x + 1$$

$$y = \frac{3}{2}x + \frac{1}{2}$$

$$m = \frac{3}{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{3}{2} - 0}{1 + \frac{3}{2} \times 0} \right|$$

$$\theta = 56^\circ 19'$$

$$\therefore \text{Angle is } 90 - 56^\circ 19' = 33^\circ$$

$$b) x^2 = 4ay$$

$$x = 2ap \quad y = ap^2$$

$$\frac{dx}{dp} = 2a \quad \frac{dy}{dp} = 2ap$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dp}}{\frac{dx}{dp}} = \frac{2ap}{2a} = p$$

$$\therefore \text{gradient of normal} = -\frac{1}{p}$$

$$\text{Now } y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

$$c) P(x) = x^3 - x^2 - 10x - 8$$

$$P(1) \neq 0, P(-1) \neq 0$$

$$P(2) \neq 0, P(-2) = 0 \therefore$$

$(x+2)$ is a factor

$$P(x) = (x+2)(x^2 - 3x - 4)$$

by inspection

$$\therefore P(x) = (x+2)(x+1)(x-4)$$

$$d) |2x - 1| = 4x + 3$$

$$2x - 1 = 4x + 3 \quad 2x - 1 = -(4x + 3)$$

$$-4 = 2x \quad 2x - 1 = -4x - 3$$

$$x = -2 \quad 6x = -2$$

$$x = -\frac{1}{3}$$

since $4x + 3 < 0$ when $x = -2$

it is no solution

$$\therefore x = -\frac{1}{3}$$

$$4 \sin(A+B) + \sin(A-B)$$

$$\sin A \cos B + \sin B \cos A + \sin A \cos B - \sin B \cos A$$

$$2 \sin A \cos B$$

$$c) 2 \sin 285^\circ \cos 45^\circ$$

$$= \sin 330^\circ + \sin 240^\circ$$

$$= -\sin 30^\circ - \sin 60^\circ$$

$$= -\frac{1}{2} - \frac{\sqrt{3}}{2}$$

$$= \frac{-1 - \sqrt{3}}{2}$$

$$b) x^2 + (k+1)x + 4$$

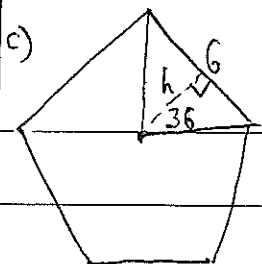
$$\Delta < 0$$

$$(k+1)^2 - 4 \times 1 \times 4 < 0$$

$$k^2 + 2k - 15 < 0$$

$$(k-3)(k+5) < 0$$

$$-5 < k < 3$$



$$\tan 36^\circ = \frac{3}{h}$$

$$h = \frac{3}{\tan 36^\circ}$$

$$= 4.129$$

$$\text{Area} = 5 \times \frac{1}{2} \times 6 \times 4.129$$

$$= 62 \text{ m}^2$$

$$d) y^2 = -3x + 6$$

$$y^2 = -3(x - 2)$$

Vertex (2, 0)

$$a = \frac{3}{4}$$

$\therefore$ Focus $(1\frac{1}{4}, 0)$

5a) $x^2 - (k+1)x + 2k+2 = 0$

b) ci) $P(x) = (x+1)(x-1)(x-3)^2(x+3)^2$

Let roots be $\alpha, 2\alpha$

$$\alpha + 2\alpha = k+1 \quad (1)$$

$$2\alpha^2 = 2k+2 \quad (2)$$

From (1), $\alpha = \frac{k+1}{3}$

sub. into (2)

$$2\left(\frac{k+1}{3}\right)^2 = 2k+2$$

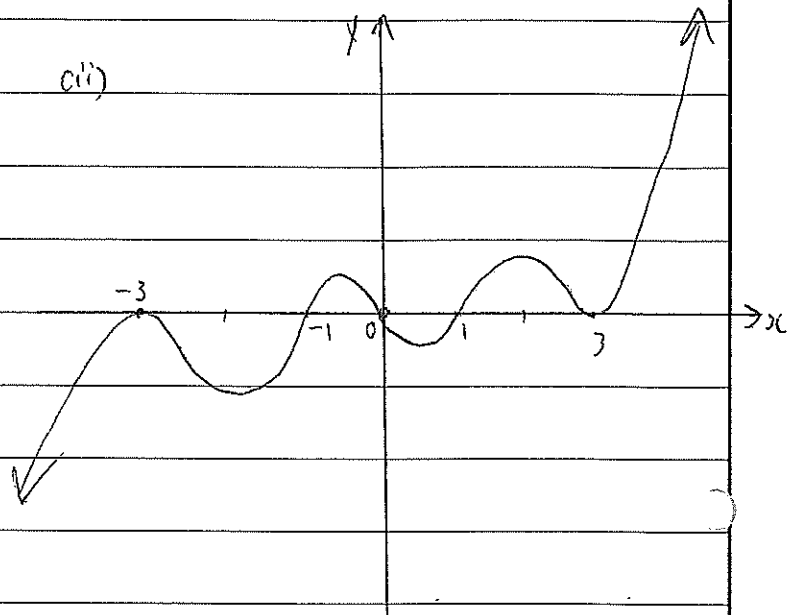
$$2k^2 + 4k + 2 = 18k + 18$$

$$2k^2 - 14k - 16 = 0$$

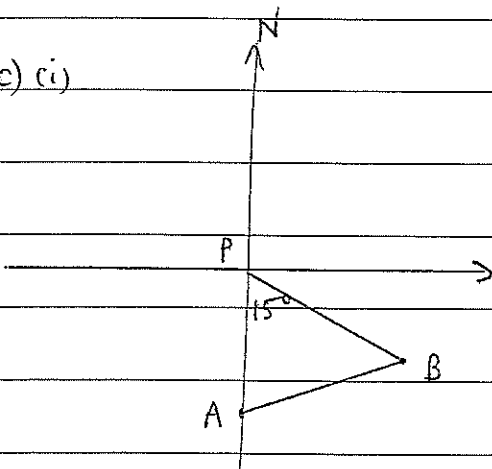
$$k^2 - 7k - 8 = 0$$

$$(k+1)(k-8) = 0$$

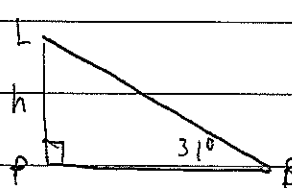
$$k = -1 \text{ or } 8$$



c) ci)

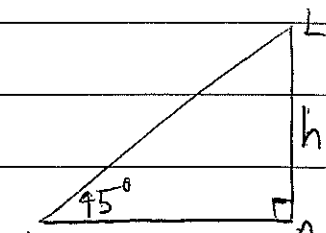


cii)



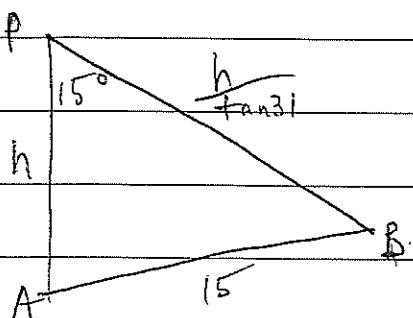
$$\tan 31^\circ = \frac{h}{BP}$$

$$BP = \frac{h}{\tan 31^\circ}$$



$$\tan 45^\circ = \frac{h}{AP}$$

$$AP = h$$



$$15^2 = h^2 + \frac{h^2}{\tan^2 31^\circ} - 2 \times \frac{h^2}{\tan 31^\circ} \times \cos 15^\circ$$

$$225 = h^2 \left(1 + \frac{1}{\tan^2 31^\circ} - \frac{2 \cos 15^\circ}{\tan 31^\circ} \right)$$

$$h^2 = 405$$

$$h = 20 \text{ m to nearest metre}$$

$$6a) P(x) = x^3 - 3x + 1$$

$$b) i) \underline{xx_0 = 4(y + y_0)}$$

Let remainder be $ax + b$

$$\therefore P(2) = 3 = 2a + b$$

$$P(-2) = -1 = -2a + b$$

$$\therefore 2 = 2b$$

$$b = 1 \therefore a = 1$$

remainder is $x + 1$

ii) Solve $x^2 = 8y$ and $xx_0 = 4(y + y_0)$ simultaneously

$$y = \frac{x^2}{8}$$

$$\therefore xx_0 = 4\left(\frac{x^2}{8} + y_0\right)$$

$$2xx_0 = x^2 + 8y_0$$

$$x^2 - 2x_0x + 8y_0 = 0$$

Now let roots be α and β

$$\therefore \alpha + \beta = 2x_0$$

$$\therefore \frac{\alpha + \beta}{2} = x_0 \text{ is } x \text{ value of midpoint}$$

Now sub. $x = x_0$ into chord of contact to get y value.

$$\frac{x_0^2}{4} = 4(y + y_0)$$

$$\frac{x_0^2}{4} = y + y_0$$

$$y = \frac{x_0^2}{4} - y_0$$

$$\therefore M \text{ is } \left(x_0, \frac{x_0^2}{4} - y_0\right)$$

cii) T lies on $x + 2y + 5 = 0$

$$\therefore x_0 + 2y_0 + 5 = 0 \quad (1)$$

Student Name: _____

Teacher Name: _____

Now for locus of M :

$$x = x_0 \quad y = \frac{x_0^2}{4} - y_0$$

$$y_0 = \frac{-x_0 - 5}{2}$$

$$\therefore y = \frac{x_0^2}{4} - \left(\frac{-x_0 - 5}{2} \right)$$

$$\therefore y = \frac{x^2}{4} + \frac{x+5}{2}$$

$$4y = x^2 + 2(x+5)$$

$$4y = x^2 + 2x + 10 \quad \text{is locus of M}$$